

Name: Vinit Mishra
PRN No: 202501040038

4.1.1. Pandas - series creation and manipulation

03:12

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the groupby and mean() operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

The user should enter a list of numbers separated by space when prompted.

Output Format:

The program should display the mean of even and odd numbers separately.

Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

```
# import pandas as pd

# # Take inputs from the user to create a list of numbers
# numbers = list(map(int, input().split()))

# grouped =

# # Display the mean of even and odd numbers with labels
# grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
# print("Mean of even and odd numbers:")
# print(grouped)

import pandas as pd

# Take inputs from the user to create a list of numbers
numbers = list(map(int, input().split()))

# Create a Pandas series from the list of numbers
s = pd.Series(numbers)

# Grouping by even and odd numbers and calculating the mean
grouped = s.groupby(s % 2 == 0).mean()

# Display the mean of even and odd numbers with labels
grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]

print("Mean of even and odd numbers:")
```

print(grouped)

Explorer

Average time
0.007 s
7.00 ms

Maximum time
0.017 s
17.00 ms

3 out of 3 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 1 **17 ms** Debug

Expected output	Actual output
1 2 3 4 5 6 7 8 9 10	1 2 3 4 5 6 7 8 9 10
Mean of even and odd numbers:	Mean of even and odd numbers:
Odd: ... 5.0	Odd: ... 5.0
Even: ... 6.0	Even: ... 6.0
dtype: float64	dtype: float64

Test case 2 **5 ms**

Test case 3 **5 ms**

Debugger

4.1.2. Dictionary to dataframe

07:26

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

Convert the dictionary to a Pandas DataFrame.

Add a new row:

Take inputs from the user for the new row data (name, age).

Add the new row to the DataFrame.

Display the DataFrame after adding the new row.

Modify a row:

Modify a specific row by changing the age. Take the row index and new age value from the user.

Display the DataFrame after modifying the row.

Delete a row:

Take the row index to be deleted from the user.

Remove the specified row.

Display the DataFrame after deleting the row.

Add a new column:

Add a column Gender with values taken from the user.

Display the DataFrame after adding the new column.

Modify a column:

Convert names to uppercase.

Display the DataFrame after modifying the column.

Delete a column:

Remove the Age column.

Display the DataFrame after deleting the column.

```
import pandas as pd
```

```
# Provided dictionary of lists
```

```
data = {  
    'Name': ['Alice', 'Bob', 'Charlie'],  
    'Age': [25, 30, 35],  
}
```

```
# Convert the dictionary to a DataFrame
```

```
df = pd.DataFrame(data)
```

```
# Display the original DataFrame
```

```
print("Original DataFrame:")
```

```
print(df)
```

```
# Adding a new row
```

```
new_name = input("New name: ")
```

```
new_age = int(input("New age: "))
```

```
df.loc[len(df)] = [new_name, new_age]
```

```
# Display the DataFrame after adding a new row
```

```
print("After adding a row:\n",df)
```

```
# Modifying a row
```

```
row_to_modify = int(input("Index of row to modify: "))
```

```
new_age_value = int(input("New age: "))
```

```
df.at[row_to_modify, "Age"] = new_age_value
```

```
# Display the DataFrame after modifying a row
```

```
print("After modifying a row:")
```

```
print(df)
```

```
# Deleting a row
```

```
row_to_delete = int(input("Index of row to delete: "))
```

```
df = df.drop(index=row_to_delete).reset_index(drop=True)
```

```
# Display the DataFrame after deleting a row
```

```
print("After deleting a row:")
```

```
print(df)
```

```
# Adding a new column
```

```
genders_input = input("Enter genders separated by space: ")
```

```
genders = genders_input.split()
```

```
# Ensure correct length
```

```
if len(genders) == len(df):
```

```
    df["Gender"] = genders
```

```
else:
```

```
    print("Error: Number of genders must match rows!")
```

```
    df["Gender"] = ["Unknown"] * len(df)
```

```
# Display the DataFrame after adding a new column
```

```
print("After adding a new column:")
```

```
print(df)
```

```
# Modifying a column
```

```
df["Name"] = df["Name"].str.upper()
```

```
# Display the DataFrame after modifying a column
```

```
print("After modifying a column:")
```

```
print(df)
```

```
# Deleting a column
```

```
df = df.drop(columns=["Age"])
```

```
# Display the DataFrame after deleting a column
```

```
print("After deleting a column:")
```


print(df)

Explorer

Average time
0.043 s
43.00 ms

Maximum time
0.049 s
49.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ 1 out of 1 hidden test case(s) passed

✓ Test case 1 49 ms Debug

Expected output

Actual output

Original DataFrame:

Original DataFrame:

.....Name...Age

.....Name...Age

0....Alice...25

0....Alice...25

1....Bob...30

1....Bob...30

2..Charlie...35

2..Charlie...35

New name: Susan

New name: Susan

New age: 29

New age: 29

After adding a row:

After adding a row:

.....Name...Age

.....Name...Age

0....Alice...25

0....Alice...25

1....Bob...30

1....Bob...30

2..Charlie...35

2..Charlie...35

3....Susan...29

3....Susan...29

Index of row to modify: 0

Index of row to modify: 0

New age: 27

New age: 27

After modifying a row:

After modifying a row:

.....Name...Age

.....Name...Age

0....Alice...27

0....Alice...27

1....Bob...30

1....Bob...30

2..Charlie...35

2..Charlie...35

3....Susan...29

3....Susan...29

Index of row to delete: 3

Index of row to delete: 3

After deleting a row:

After deleting a row:

.....Name...Age

.....Name...Age

0....Alice...27

0....Alice...27

1....Bob...30

1....Bob...30

.....

.....

Terminal Test cases

Average time

0.043 s

43.00 ms

Maximum time

0.049 s

49.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ 1 out of 1 hidden test case(s) passed

After modifying a row:

.....Name..Age

0....Alice...27

1....Bob...30

2..Charlie...35

3....Susan...29

Index of row to delete: 3

After deleting a row:

.....Name..Age

0....Alice...27

1....Bob...30

2..Charlie...35

Enter genders separated by space: F M M

After adding a new column:

.....Name..Age..Gender

0....Alice...27.....F

1....Bob...30.....M

2..Charlie...35.....M

After modifying a column:

.....Name..Age..Gender

0....ALICE...27.....F

1....BOB...30.....M

2..CHARLIE...35.....M

After deleting a column:

.....Name..Gender

0....ALICE.....F

1....BOB.....M

2..CHARLIE.....M

After modifying a row:

.....Name..Age

0....Alice...27

1....Bob...30

2..Charlie...35

3....Susan...29

Index of row to delete: 3

After deleting a row:

.....Name..Age

0....Alice...27

1....Bob...30

2..Charlie...35

Enter genders separated by space: F M M

After adding a new column:

.....Name..Age..Gender

0....Alice...27.....F

1....Bob...30.....M

2..Charlie...35.....M

After modifying a column:

.....Name..Age..Gender

0....ALICE...27.....F

1....BOB...30.....M

2..CHARLIE...35.....M

After deleting a column:

.....Name..Gender

0....ALICE.....F

1....BOB.....M

2..CHARLIE.....M

Terminal

Test cases

4.1.3. Student Information

01:49

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

Display the first five rows of the data frame.

Calculate the average age of the students (limit the average age up to 2 decimal places).

Filter out the students who have a grade above a certain threshold (consider the threshold grade is **'B'**).

Note:

Refer to the displayed test cases for better understanding.

```
import pandas as pd
```

```
# Read the text file into a DataFrame
file = input()
data = pd.read_csv(file, sep="\s+", header=None, names=["Name", "Age", "Grade"])
# Display the first five rows
print("First five rows:")
print(data.head())

# Calculate the average age (2 decimal places)
avg_age = round(data["Age"].mean(), 2)
print("Average age:", avg_age)

# Filter students with grade up to 'B' (A and B)
filtered = data[data["Grade"].isin(['A', 'B'])]

print("Students with a grade up to B")
```

print(filtered)

Average time
0.028 s
27.50 ms

Maximum time
0.040 s
40.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ 1 out of 1 hidden test case(s) passed

✓ Test case 1 40 ms Debug

Expected output

Actual output

studentdata.txt

First five rows:

...

0 John 25 A

1 Alin 22 B

2 Emma 24 A

3 Mike 23 C

4 Sony 21 B

Average age: 23.29

Students with a grade up to B

...

0 John 25 A

1 Alin 22 B

2 Emma 24 A

4 Sony 21 B

5 Amia 26 A

studentdata.txt

First five rows:

...

0 John 25 A

1 Alin 22 B

2 Emma 24 A

3 Mike 23 C

4 Sony 21 B

Average age: 23.29

Students with a grade up to B

...

0 John 25 A

1 Alin 22 B

2 Emma 24 A

4 Sony 21 B

5 Amia 26 A

Terminal

Test cases

< Prev

Reset

Submit

Next >

4.1.3. Student Information

01:49

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

Display the first five rows of the data frame.

Calculate the average age of the students (limit the average age up to 2 decimal places).

Filter out the students who have a grade above a certain threshold (consider the threshold grade is **'B'**).

Note:

Refer to the displayed test cases for better understanding.

```
import pandas as pd
```

```
# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)

# Prepare data for calculation
df['Date'] = pd.to_datetime(df['Date'])
df['Month'] = df['Date'].dt.to_period('M')
df['Sales'] = df['Quantity'] * df['Price']

# Find the month with the highest total sales
best_month = df.groupby('Month')['Sales'].sum().idxmax()
highest_sales = df.groupby('Month')['Sales'].sum().max()

print(f"Best month: {best_month}")
print(f"Total sales: ${highest_sales:.2f}")
```

The screenshot displays a test runner interface with the following components:

- Performance Metrics:**
 - Average time: 0.035 s (34.67 ms)
 - Maximum time: 0.063 s (63.00 ms)
- Test Results Summary:**
 - 1 out of 1 shown test case(s) passed
 - 2 out of 2 hidden test case(s) passed
- Test Case Details (Test case 1):**
 - Status: 63 ms, with a 'Debug' button.
 - Expected output:**
 - sales_data.csv
 - Best month: 2025-01
 - Total sales: \$1210.00
 - Actual output:**
 - sales_data.csv
 - Best month: 2025-01
 - Total sales: \$1210.00
- Navigation:** Buttons for 'Terminal' and 'Test cases' are at the bottom.

4.2.2. Best Selling Product

02:08

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the columns: Date, Product, Quantity, Price, and City.

Find the product that sold the most in terms of quantity sold.

Display the product that sold the most and the total quantity sold for that product.

Sample Data :

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los Angeles

2025-01-05,Product A,3,20,Chicago

2025-01-05,Product C,10,30,Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

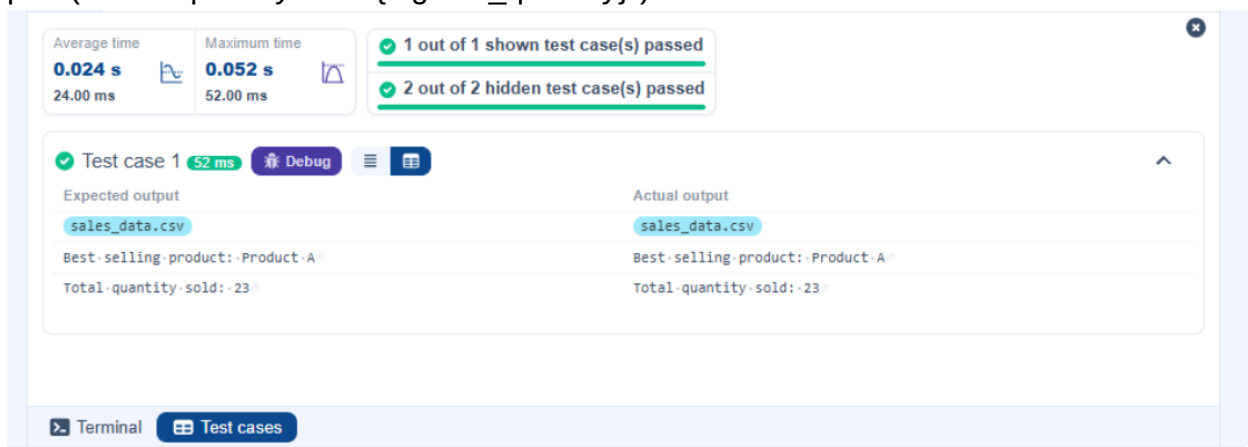

```
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)

# Find the product with the highest total quantity sold
best_product = df.groupby('Product')['Quantity'].sum().idxmax()
highest_quantity = df.groupby('Product')['Quantity'].sum().max()

# Display the result
print(f"Best selling product: {best_product}")
print(f"Total quantity sold: {highest_quantity}")
```



The screenshot displays a code execution interface with the following components:

- Performance Metrics:**
 - Average time: 0.024 s (24.00 ms)
 - Maximum time: 0.052 s (52.00 ms)
- Test Results Summary:**
 - 1 out of 1 shown test case(s) passed
 - 2 out of 2 hidden test case(s) passed
- Test Case Details (Test case 1):**
 - Status: Passed (52 ms)
 - Buttons: Debug, Menu, Expand
 - Expected output:**
 - sales_data.csv
 - Best-selling product: Product-A
 - Total quantity sold: 23
 - Actual output:**
 - sales_data.csv
 - Best-selling product: Product-A
 - Total quantity sold: 23
- Navigation:** Terminal, Test cases

4.2.3. City that Sold the Most Products

01:45

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the columns: Date, Product, Quantity, Price, and City.

Group the data by City and calculate the total quantity of products sold for each city.

Find the city that sold the most products (based on the total quantity sold).

Sample Data :

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los Angeles

2025-01-05,Product A,3,20,Chicago

2025-01-05,Product C,10,30,Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)

# Group by City and calculate total quantity
city_sales = df.groupby('City')['Quantity'].sum()

# Find the city with the highest total quantity sold
best_city = city_sales.idxmax()

# Display the result
print(f"City sold the most products: {best_city}")
```

The screenshot displays a code execution interface with the following components:

- Performance Metrics:**
 - Average time: 0.018 s (18.33 ms)
 - Maximum time: 0.036 s (36.00 ms)
- Test Results:**
 - 1 out of 1 shown test case(s) passed
 - 2 out of 2 hidden test case(s) passed
- Test Case Details:**
 - Test case 1:** 36 ms, with a 'Debug' button.
 - Expected output:** sales_data.csv, City sold the most products: Los Angeles
 - Actual output:** sales_data.csv, City sold the most products: Los Angeles
- Navigation:** Buttons for 'Terminal' and 'Test cases' at the bottom.

4.2.4. Most Frequently Sold Product Pairs

01:25

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the following columns: Date, Product, Quantity, Price, and City.

For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).

Output the product pair/s that was sold most frequently.

Sample Data :

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los Angeles

2025-01-05,Product A,3,20,Chicago

2025-01-05,Product C,10,30,Los Angeles

Explanation:

Transactions:

2025-01-01: Product A, Product B

2025-01-02: Product A, Product C

2025-01-03: Product B, Product A

2025-01-04: Product C, Product B

2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together:

Product A and Product B : Appear in transactions on 2025-01-01 and 2025-01-03.

Product A and Product C : Appear in transactions on 2025-01-02 and 2025-01-05.

Product B and Product C : Appears in transactions on 2025-01-04.

Most Frequent Product Combinations:

Product A and Product B (2 times)

Product A and Product C (2 times)

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```
import pandas as pd
```

```
from itertools import combinations
from collections import Counter

# Prompt user to input the file name
file_name = input()

# Read data from the specified CSV file
df = pd.read_csv(file_name)

# write the code
pair_counter = Counter()

# Group by Date and generate product pairs
for _, group in df.groupby('Date'):
    products = group['Product'].tolist()
    pairs = combinations(sorted(products), 2)
    pair_counter.update(pairs)

# Find maximum frequency
max_count = max(pair_counter.values())

# Output the most frequent product pairs
for pair, count in pair_counter.items():
    if count == max_count:
```

```
print(f"{pair[0]} and {pair[1]}: {count} times")
```

Average time
0.020 s
20.33 ms

Maximum time
0.039 s
39.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 39 ms Debug

Expected output

Actual output

sales_data.csv

sales_data.csv

Product A and Product B: 2 times

Product A and Product B: 2 times

Product A and Product C: 2 times

Product A and Product C: 2 times

Terminal

Test cases

4.2.5. Titanic Dataset Analysis and Data Cleaning

01:36

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using `.info()`).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.
8. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
9. Drop the 'Cabin' column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina
Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# 1. Display the first 5 rows of the dataset
print(data.head())

# 2. Display the last 5 rows of the dataset
print(data.tail())

# 3. Get the shape of the dataset
print(data.shape)

# 4. Get a summary of the dataset (info)
print(data.info())

# 5. Get basic statistics of the dataset
print(data.describe())

# 6. Check for missing values
print(data.isnull().sum())

# 7. Fill missing values in the 'Age' column with the median age
data['Age'] = data['Age'].fillna(data['Age'].median())
```

8. Fill missing values in the 'Embarked' column with the mode

```
data['Embarked'] = data['Embarked'].fillna(data['Embarked'].mode()[0])
```

9. Drop the 'Cabin' column due to many missing values

```
data = data.drop(columns=['Cabin'])
```

10. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch'

```
data['FamilySize'] = data['SibSp'] + data['Parch']
```

Average time 0.166 s
Maximum time 0.166 s
1 out of 1 shown test case(s) passed

Test case 1 166 ms Debug

Expected output	Actual output
PassengerId Survived Pclass Fare Cabin Embarked	PassengerId Survived Pclass Fare Cabin Embarked
0 1 0 3 7.2500 NaN S	0 1 0 3 7.2500 NaN S
1 2 1 1 71.2833 C85 C	1 2 1 1 71.2833 C85 C
2 3 1 3 7.9250 NaN S	2 3 1 3 7.9250 NaN S
3 4 1 1 53.1000 C123 S	3 4 1 1 53.1000 C123 S
4 5 0 3 8.0500 NaN S	4 5 0 3 8.0500 NaN S
[5 rows x 12 columns]	[5 rows x 12 columns]
PassengerId Survived Pclass Fare Cabin Embarked	PassengerId Survived Pclass Fare Cabin Embarked
886 887 0 2 13.00 NaN S	886 887 0 2 13.00 NaN S
887 888 1 1 30.00 B42 S	887 888 1 1 30.00 B42 S
888 889 0 3 23.45 NaN S	888 889 0 3 23.45 NaN S
889 890 1 1 30.00 C148 C	889 890 1 1 30.00 C148 C
890 891 0 3 7.75 NaN Q	890 891 0 3 7.75 NaN Q

Average time

0.166 s

166.00 ms



Maximum time

0.166 s

166.00 ms



1 out of 1 shown test case(s) passed

```
[5 rows x 12 columns]
(891, 12)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   PassengerId      891 non-null    int64
1   Survived         891 non-null    int64
2   Pclass           891 non-null    int64
3   Name             891 non-null    object
4   Sex              891 non-null    object
5   Age              714 non-null    float64
6   SibSp            891 non-null    int64
7   Parch            891 non-null    int64
8   Ticket           891 non-null    object
9   Fare             891 non-null    float64
10  Cabin            204 non-null    object
11  Embarked         889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 66.2+ KB
None
.....PassengerId.....Survived.....Pclass.....SibSp.....Parch.....Fare
count.....891.000000.....891.000000.....891.000000.....891.000000.....891.000000.....891.000000
mean.....446.000000.....0.383838.....2.308642.....0.523008.....0.381594.....32.204208
std.....257.353842.....0.486592.....0.836071.....1.102743.....0.806057.....49.693429

[5 rows x 12 columns]
(891, 12)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   PassengerId      891 non-null    int64
1   Survived         891 non-null    int64
2   Pclass           891 non-null    int64
3   Name             891 non-null    object
4   Sex              891 non-null    object
5   Age              714 non-null    float64
6   SibSp            891 non-null    int64
7   Parch            891 non-null    int64
8   Ticket           891 non-null    object
9   Fare             891 non-null    float64
10  Cabin            204 non-null    object
11  Embarked         889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 66.2+ KB
None
.....PassengerId.....Survived.....Pclass.....SibSp.....Parch.....Fare
count.....891.000000.....891.000000.....891.000000.....891.000000.....891.000000.....891.000000
mean.....446.000000.....0.383838.....2.308642.....0.523008.....0.381594.....32.204208
std.....257.353842.....0.486592.....0.836071.....1.102743.....0.806057.....49.693429
```

Terminal

Test cases

Navigation icons: Home, Back, Forward, Search, and a taskbar with icons for File Explorer, Edge, and Word. System tray shows time 9:37 PM, date 4/24/2026, and status icons for network, volume, and battery.

4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

02:18

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
2. Convert the 'Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the 'Embarked' column, dropping the first category.
4. Get the mean age of passengers.
5. Get the median fare of passengers.
6. Get the number of passengers by class.
7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']
# 1. Create a new column 'IsAlone' (1 if alone, 0 otherwise)
data['IsAlone'] = np.where(data['FamilySize'] == 0, 1, 0)

# 2. Convert 'Sex' to numeric (male: 0, female: 1)
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

# 3. One-hot encode the 'Embarked' column
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# 4. Get the mean age of passengers
print(data['Age'].mean())

# 5. Get the median fare of passengers
print(data['Fare'].median())

# 6. Get the number of passengers by class
print(data['Pclass'].value_counts())

# 7. Get the number of passengers by gender
print(data['Sex'].value_counts())
```


8. Get the number of passengers by survival status

```
print(data['Survived'].value_counts())
```

9. Calculate the survival rate

```
print(data['Survived'].mean())
```

10. Calculate the survival rate by gender

```
print(data.groupby('Sex')['Survived'].mean())
```

Average time: 0.079 s (79.00 ms) | Maximum time: 0.079 s (79.00 ms) | 1 out of 1 shown test case(s) passed

Test case 1 (79 ms) [Debug] [Menu] [Calendar]

Expected output	Actual output
29.69911764705882	29.69911764705882
14.4542	14.4542
3...491	3...491
1...216	1...216
2...184	2...184
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64
0...577	0...577
1...314	1...314
Name: Sex, dtype: int64	Name: Sex, dtype: int64
0...549	0...549
1...342	1...342
Name: Survived, dtype: int64	Name: Survived, dtype: int64
0.3838383838383838	0.3838383838383838
Sex	Sex
0...0.188908	0...0.188908
1...0.742038	1...0.742038
Name: Survived, dtype: float64	Name: Survived, dtype: float64

4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

03:10

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
```

```
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']
data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# 1. Calculate the survival rate by class
print(data.groupby('Pclass')['Survived'].mean())

# 2. Calculate the survival rate by embarked location
print(data.groupby('Embarked_S')['Survived'].mean())

# 3. Calculate the survival rate by family size
print(data.groupby('FamilySize')['Survived'].mean())

# 4. Calculate the survival rate by being alone
print(data.groupby('IsAlone')['Survived'].mean())

# 5. Get the average fare by class
print(data.groupby('Pclass')['Fare'].mean())

# 6. Get the average age by class
print(data.groupby('Pclass')['Age'].mean())

# 7. Get the average age by survival status
print(data.groupby('Survived')['Age'].mean())
```

8. Get the average fare by survival status

```
print(data.groupby('Survived')['Fare'].mean())
```

9. Get the number of survivors by class

```
print(data[data['Survived'] == 1]['Pclass'].value_counts())
```

10. Get the number of non-survivors by class

```
print(data[data['Survived'] == 0]['Pclass'].value_counts())
```

Explorer

Debugger

Average time
0.103 s
103.00 ms

Maximum time
0.103 s
103.00 ms

1 out of 1 shown test case(s) passed

Test case 1 103 ms Debug

Expected output

Actual output

Pclass	Pclass
1...0.629630	1...0.629630
2...0.472826	2...0.472826
3...0.242363	3...0.242363
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Embarked_S	Embarked_S
0...0.506073	0...0.506073
1...0.336957	1...0.336957
Name: Survived, dtype: float64	Name: Survived, dtype: float64
FamilySize	FamilySize
0...0.303538	0...0.303538
1...0.552795	1...0.552795
2...0.578431	2...0.578431
3...0.724138	3...0.724138
4...0.200000	4...0.200000
5...0.136364	5...0.136364
6...0.333333	6...0.333333
7...0.000000	7...0.000000
10...0.000000	10...0.000000
Name: Survived, dtype: float64	Name: Survived, dtype: float64
IsAlone	IsAlone
0...0.505650	0...0.505650
1...0.303538	1...0.303538
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Pclass	Pclass
1...84.154687	1...84.154687
2...20.662183	2...20.662183
3...13.675550	3...13.675550
Name: Fare, dtype: float64	Name: Fare, dtype: float64
Pclass	Pclass
1...38.233441	1...38.233441
2...29.877630	2...29.877630

Average time
0.103 s
103.00 ms



Maximum time
0.103 s
103.00 ms



1 out of 1 shown test case(s) passed



5....0.136364	5....0.136364
6....0.333333	6....0.333333
7....0.000000	7....0.000000
10....0.000000	10....0.000000
Name: Survived, dtype: float64	Name: Survived, dtype: float64
IsAlone	IsAlone
0....0.505650	0....0.505650
1....0.303538	1....0.303538
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Pclass	Pclass
1....84.154687	1....84.154687
2....20.662183	2....20.662183
3....13.675550	3....13.675550
Name: Fare, dtype: float64	Name: Fare, dtype: float64
Pclass	Pclass
1....38.233441	1....38.233441
2....29.877630	2....29.877630
3....25.140620	3....25.140620
Name: Age, dtype: float64	Name: Age, dtype: float64
Survived	Survived
0....30.626179	0....30.626179
1....28.343690	1....28.343690
Name: Age, dtype: float64	Name: Age, dtype: float64
Survived	Survived
0....22.117887	0....22.117887
1....48.395408	1....48.395408
Name: Fare, dtype: float64	Name: Fare, dtype: float64
1....136	1....136
3....119	3....119
2....87	2....87
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64
3....372	3....372
2....97	2....97
1....80	1....80